Practical No:1 -B

Aim: Develop an Android application demonstrating the use of Programming Resources (colors.xml, themes.xml, strings.xml, dimens.xml, drawable resources) and apply them in UI components.

Android provides resource files to separate UI data from code, making applications easier to manage and maintain.

Resource Files

- strings.xml → Stores text values.
- colors.xml → Stores color values.
- dimens.xml → Stores dimensions such as text size and margins.
- themes.xml → Defines the application's theme.
- drawable → Stores images, shapes, and backgrounds.

Define following colors in Color.xml code:-

Go to res-values-color.xml and open the color.xml file and add the following colors to it.

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFFF</color>
    <color name="Red">#C30A0A</color>
    <color name="Green">#23BF29</color>
    <color name="Purple">#CB4FE0</color>
</resources>
```

Activicty_main.xml code:-

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"

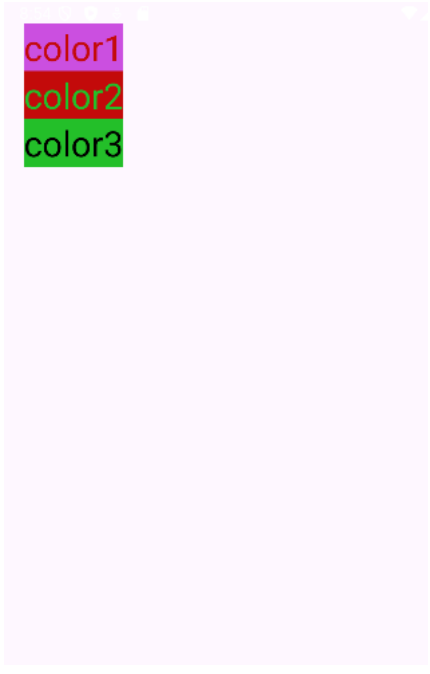
    android:orientation="vertical"
    android:padding="20dp">

    <TextView
        android:id="@+id/textView2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="color1"
         android:textColor="@color/Red"
        android:textSize="30sp"
        android:background="@color/Purple"/>
```

```
    <TextView
        android:id="@+id/textView4"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:background="@color/Red"
        android:text="color2"
        android:textColor="@color/Green"
        android:textSize="30sp"/>

    <TextView
        android:id="@+id/textView3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textColor="@color/black"
        android:background="@color/Green"
        android:text="color3"
        android:textSize="30sp"/>
</LinearLayout>
```

Output:-



2) Theme

Go to res-values-themes and open thems.xml file

themes.xml code:

activity_main.xml | themes.xml | colors.xml | MainActivity.java

```
<resources xmlns:tools="http://schemas.android.com/tools">
    <!-- Base application theme. -->
    <style name="Base.Theme.Audio" parent="Theme.Material3.DayNight.NoActionBar">
        <!-- Customize your light theme here. -->
        <!-- <item name="colorPrimary">@color/my_light_primary</item> -->
    </style>

    <style name="Theme.Audio" parent="Base.Theme.Audio" />
</resources>
```

activity_main.xml | themes.xml | colors.xml | MainActivity.java

```
<resources xmlns:tools="http://schemas.android.com/tools">
    <!-- Base application theme. -->
    <style name="Base.Theme.Audio" parent="Theme.Material3.Dark">
        <!-- Customize your light theme here. -->
        <!-- <item name="colorPrimary">@color/my_light_primary</item> -->
    </style>

    <style name="Theme.Audio" parent="Base.Theme.Audio" />
</resources>
```

```
<resources xmlns:tools="http://schemas.android.com/tools">
    <!-- Base application theme. -->
    <style name="Base.Theme.Audio" parent="Theme.Material3.DayNight">
        <!-- Customize your light theme here. -->
        <!-- <item name="colorPrimary">@color/my_light_primary</item> -->
    </style>

    <style name="Theme.Audio" parent="Base.Theme.Audio" />
</resources>
    </resources>
```



3) String

Go to res-values-string and open string.xml file

Write following code to string.xml file

String.xml code:-

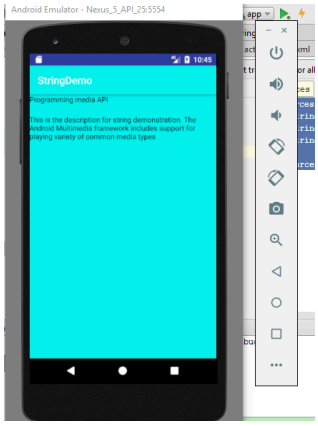
```
<resources>
<string name="app_name">StringDemo</string>
<string name="ParaHead">Programming media API</string>
<string name="Description">This is the description for string demonstration. The Android Multimedia framework includes support for playing variety of common media types</string>
</resources>
```

Activity_main.xml code:-

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
```

```
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context="com.example.student.pract1.MainActivity"
    android:layout_weight="3">
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="@string/ParaHead"/>
<TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/Description" android:layout_marginTop="20dp" />
</LinearLayout>
```

Output:-



4) Drawables and images.

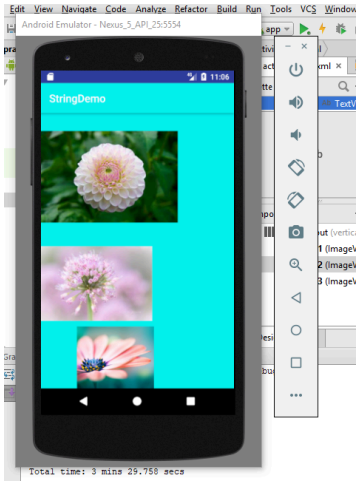
Copy and Paste the three images in the drawable folder.

Activity_main.xml code:-

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context="com.example.student.pract1.MainActivity">
<ImageView
    android:layout_width="256dp"
```

```
    android:layout_height="237dp"
    android:id="@+id/image1"
    android:src="@drawable/image1"/>
<ImageView
        android:id="@+id/image2"
        android:layout_width="208dp"
        android:layout_height="160dp"
        android:src="@drawable/image2" />
<ImageView
        android:id="@+id/image3"
        android:layout_width="279dp"
        android:layout_height="wrap_content"
        android:src="@drawable/image3" />
</LinearLayout>
```

Output:- Drawable



5) Dimension

Dimens.xml

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
<dimen name="textview_height">35dp</dimen>
<dimen name="textview_width">150dp</dimen>
<dimen name="font_size">26sp</dimen>
</resources>
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
```

```
    android:orientation="vertical"
    tools:context="com.example.student.pract1.MainActivity">
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Normal Text"/>
<TextView
        android:layout_width="@dimen/textview_width"
        android:layout_height="@dimen/textview_height"
        android:textSize="@dimen/font_size"
        android:text="Modified dimension Text"
/>
</LinearLayout>
```

Output:-

